

TuniCars+

Resilience Testing Report

Docker Swarm vs Kubernetes

Session ID

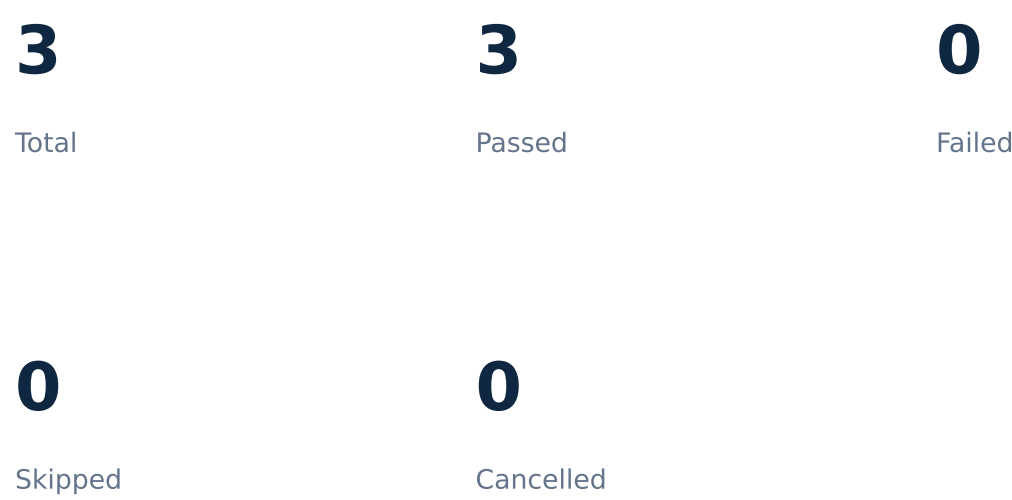
20260813_172814_d7fe25

Date

2026-08-13T16:28:14.697446+00:00

Executive Summary

20260813_172814_d7fe25



Overall measured availability: 100.00%

During this campaign, 3 of 3 scenarios passed, 0 failed and 0 were skipped. Conclusions are limited to the measured scenarios and recorded environment constraints.

Test Environment

Architecture and technologies

Frontend → Backend → PostgreSQL

Docker Swarm environment

- Docker services
- FastAPI backend
- Next.js frontend
- PostgreSQL database

Kubernetes / Minikube environment

- Kubernetes Deployments and Services
- Minikube nodes
- FastAPI · Next.js · PostgreSQL

Test Scenarios

Scenario	kubernetes
cpu	3

Full Results Table

Scenario	Platform	Detection	Recovery	Required	Success	Avg ms	P95 ms	HTTP Fail	Avail. %	Status	Skip Reason	Error Message
cpu	kubernetes	N/A	N/A	N/A	Not Required	20.911	32.364	N/A	100.0	PASS	N/A	N/A
cpu	kubernetes	N/A	N/A	N/A	Not Required	20.846	34.939	N/A	100.0	PASS	N/A	N/A
cpu	kubernetes	N/A	N/A	N/A	Not Required	19.209	29.349	N/A	100.0	PASS	N/A	N/A

kubernetes — cpu

Status: Pass
Detection Time (s): N/A
Recovery Time (s): N/A
HTTP Success: N/A
HTTP Failures: N/A
Availability (%): 100.0
CPU (%): N/A
Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

kubernetes — cpu

Status: Pass
Detection Time (s): N/A
Recovery Time (s): N/A
HTTP Success: N/A
HTTP Failures: N/A
Availability (%): 100.0
CPU (%): N/A
Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

kubernetes — cpu

Status: Pass
Detection Time (s): N/A
Recovery Time (s): N/A
HTTP Success: N/A
HTTP Failures: N/A
Availability (%): 100.0
CPU (%): N/A
Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

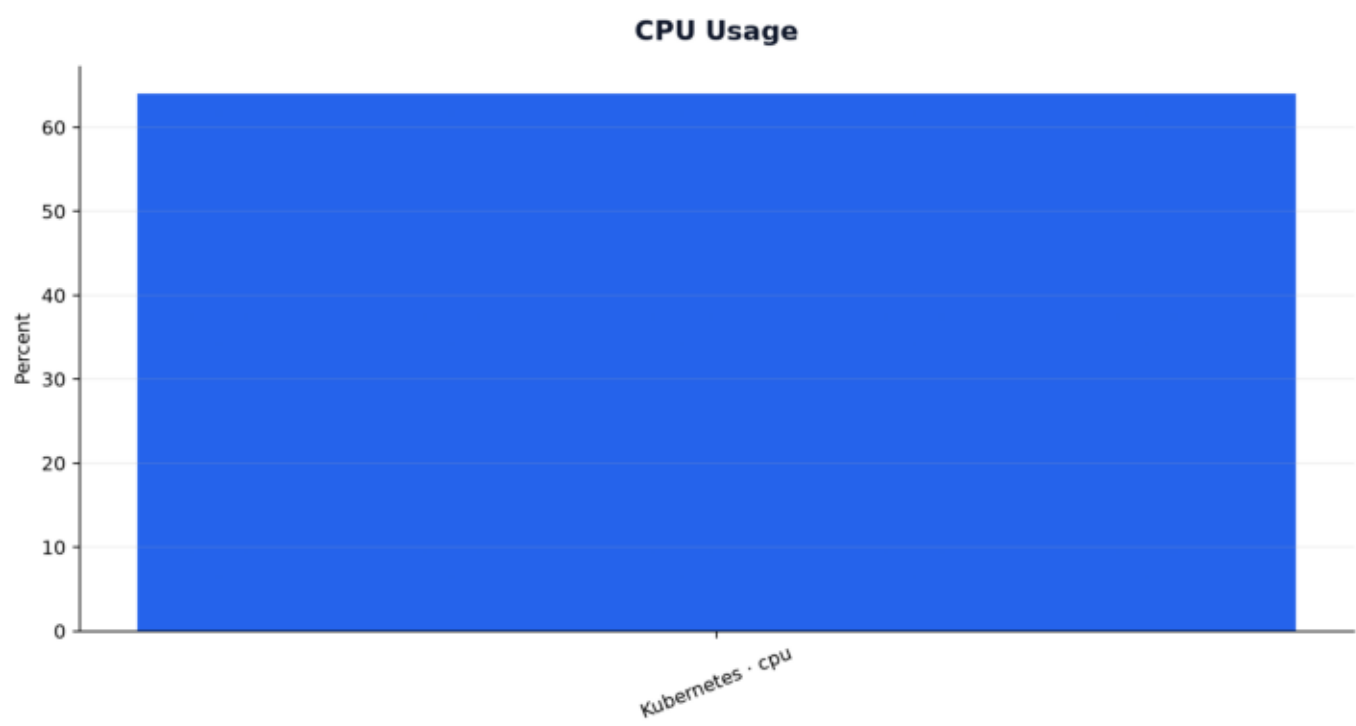
Availability

Current-session measured data



Cpu Usage

Current-session measured data



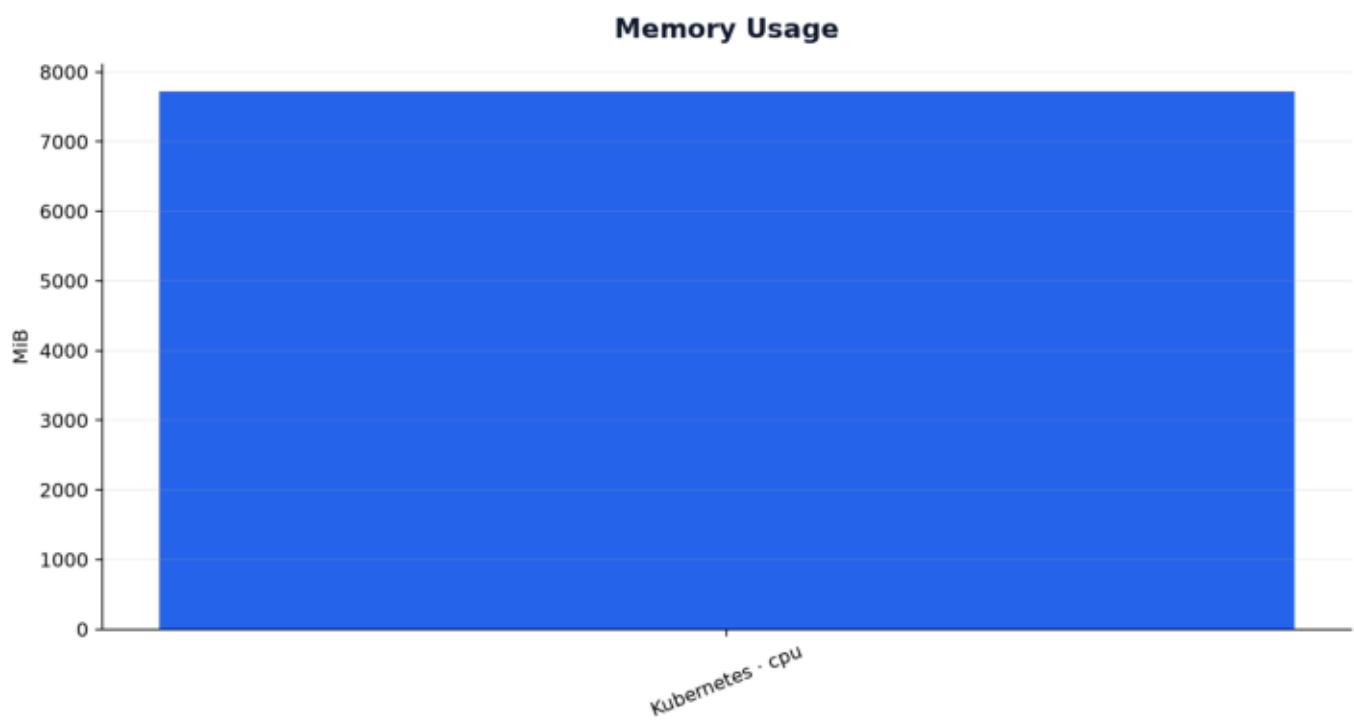
Http Failures

Current-session measured data



Memory Usage

Current-session measured data



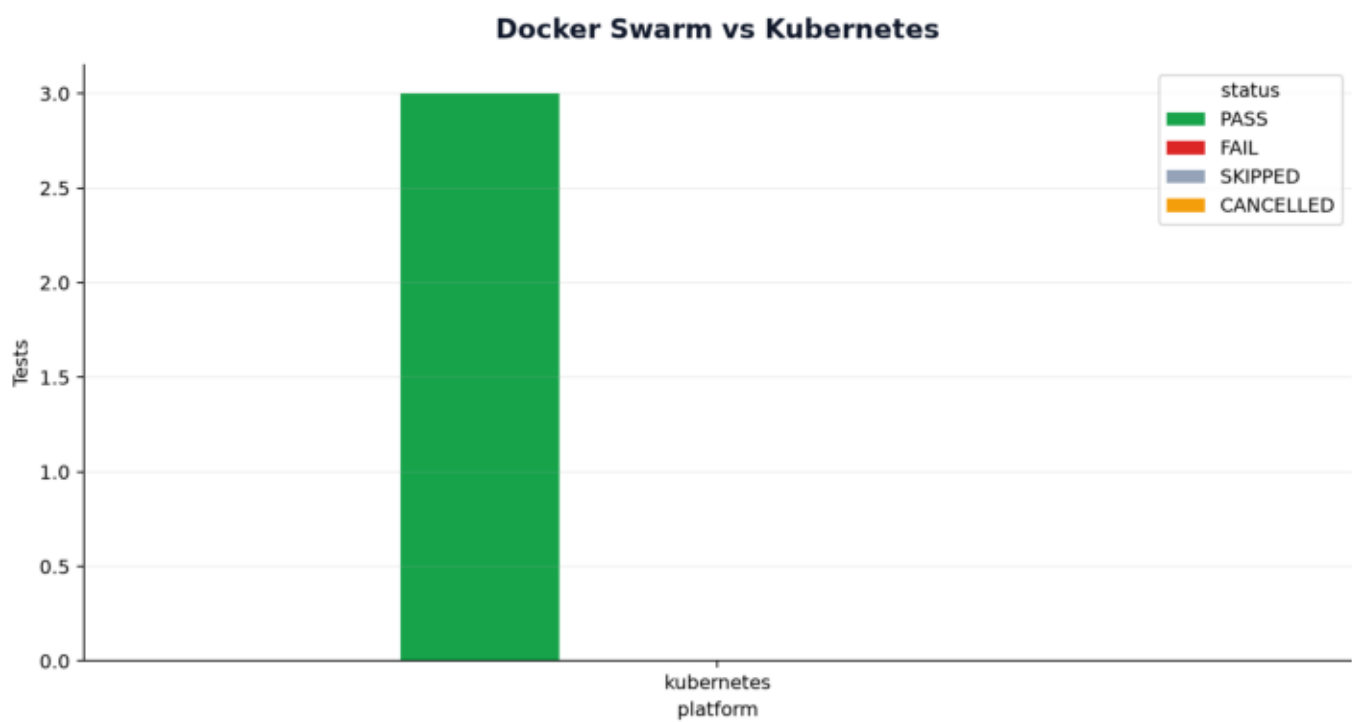
P95 Response Time

Current-session measured data



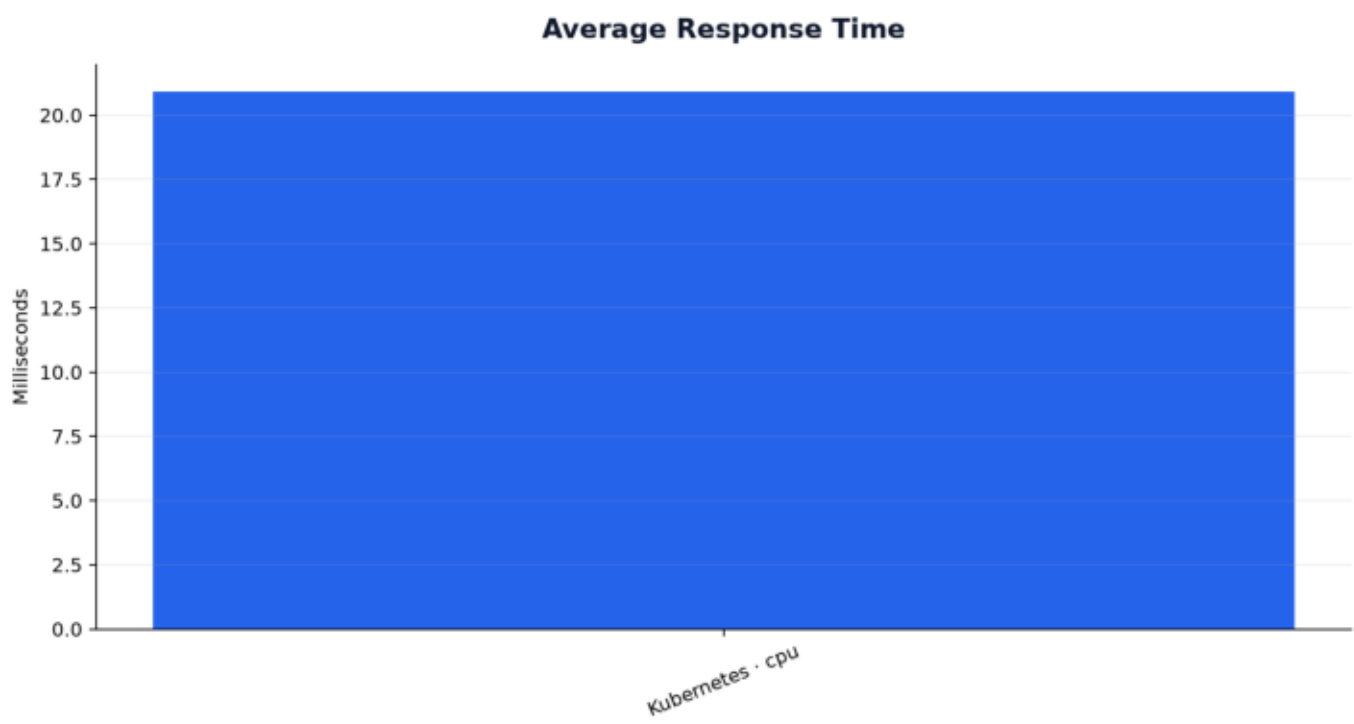
Platform Comparison

Current-session measured data



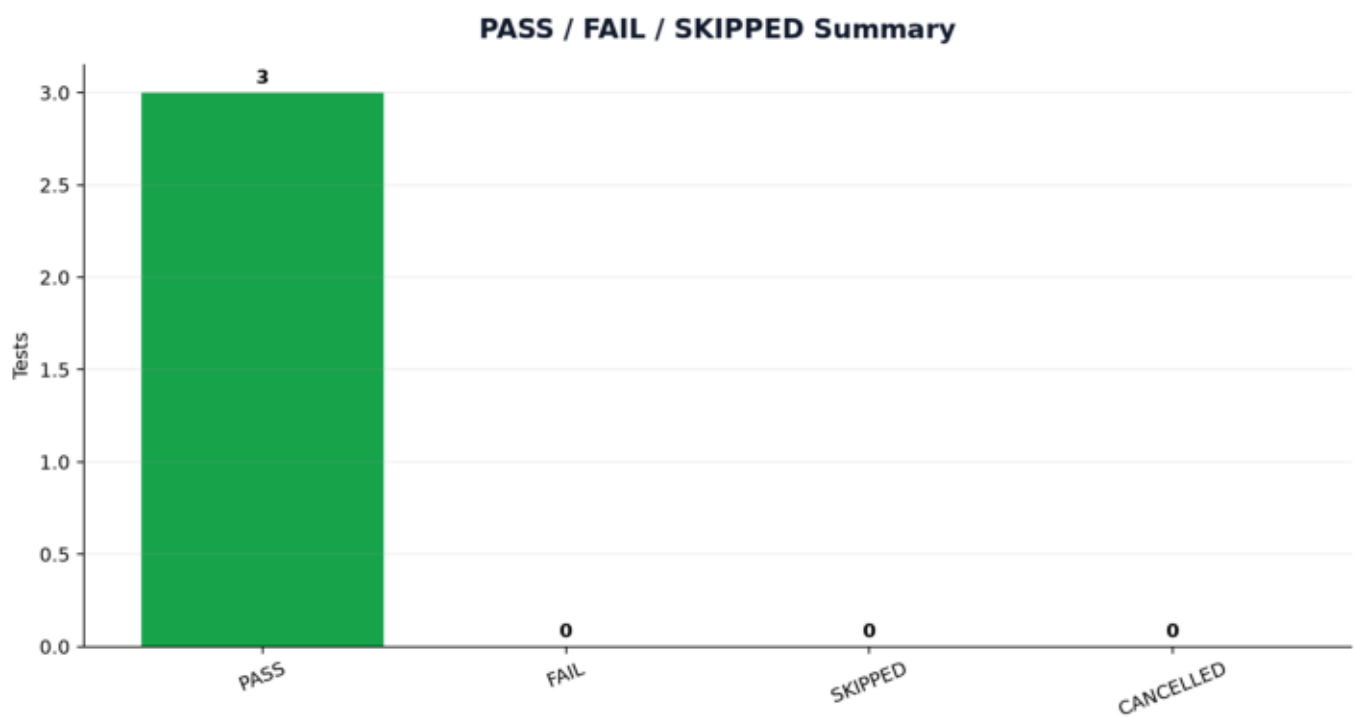
Response Time

Current-session measured data



Status Summary

Current-session measured data



Failures and Errors

Platform	Scenario	Status	Error
—	—	—	No failures recorded.

Skipped Tests

Platform	Scenario	Exact reason
—	—	No skipped tests recorded.

Swarm vs Kubernetes Comparison

Platform	PASS
kubernetes	3

Final Conclusion

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During this campaign, 3 of 3 scenarios passed, 0 failed and 0 were skipped. Conclusions are limited to the measured scenarios and recorded environment constraints.